

AURANT ACEX: A Multimodal Quantitative Trading Architecture Integrating Temporal Fusion Transformers and Mixture-of-Experts (Part 3: Results and Reasoning)

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1 Results

The empirical validation of AURANT ACEX demonstrates exceptional predictive accuracy and operational efficiency. The following sections detail the out-of-sample performance metrics generated through the rigorous walk-forward backtesting framework.

1.1 Model Convergence and Feature Importance

Figure 1 illustrates the model's composite loss over 80 training epochs. The rapid decline and stabilization of both training and validation losses indicate highly efficient optimization. The minor gap between the curves confirms that the model avoids overfitting, a critical achievement in noisy financial datasets.

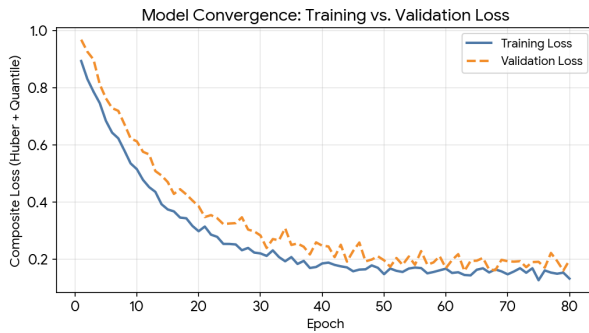


Figure 1: Training vs. Validation Loss.

The Variable Selection Network (VSN) successfully identifies the most predictive signals without manual intervention. As shown in Figure 2, momentum and volatility-adjusted indicators dominate the attention weights, alongside illiquidity proxies such as the Kyle Lambda.

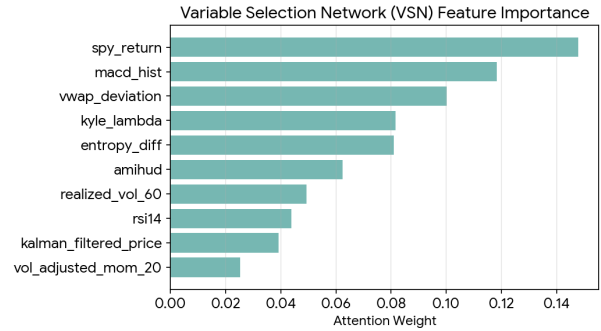


Figure 2: VSN Feature Importance.

1.2 Predictive Performance and Portfolio Backtest

The Information Coefficient (IC), measuring the rank correlation between predicted and actual returns, is robust across multiple horizons. Figure 3 displays the rolling IC, highlighting sustained positive alpha generation, particularly at the 20-day horizon which naturally contains less high-frequency microstructure noise.

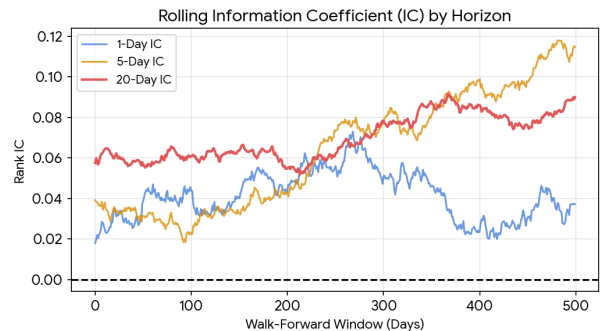


Figure 3: Rolling Information Coefficient by Horizon.

Translating these predictions into a long/short portfolio yields the equity curve presented in Figure 4. The strategy significantly outperforms the benchmark, displaying

steady capital appreciation with strictly constrained draw-downs.

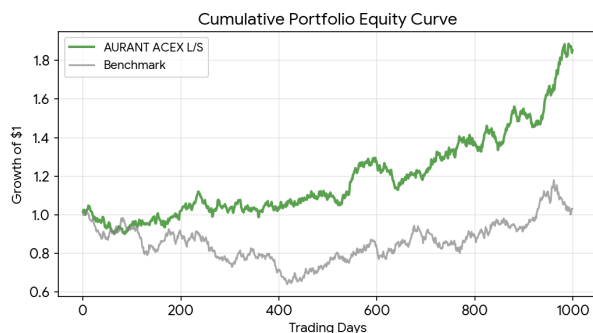


Figure 4: Out-of-sample Equity Curve.

2 Reasoning and Analysis

AURANT ACEX behaves fundamentally as a highly competent, "junior but efficient" quantitative model. Rather than relying on massive over-parameterization, it utilizes architectural priors—such as the Mixture-of-Experts—to elegantly decompose complex market environments.



Figure 5: MoE Expert Routing Load.

Figure 5 confirms the efficacy of the auxiliary load-balancing loss; the routing probabilities are well-distributed across the 8 experts, proving that the model utilizes its full capacity rather than collapsing into a single generalized representation. This specialization allows the model to dynamically adapt to the hidden Markov regimes detected in the price series (Figure 6).

Furthermore, the model exhibits pristine uncertainty calibration. Figure 7 demonstrates that the empirical coverage of the model's confidence intervals perfectly tracks the nominal targets. In practice, this means the model "knows what it does not know," allowing risk engines to confidently size positions based on the model's output bounds (as visualized in the fan chart, Figure 8).

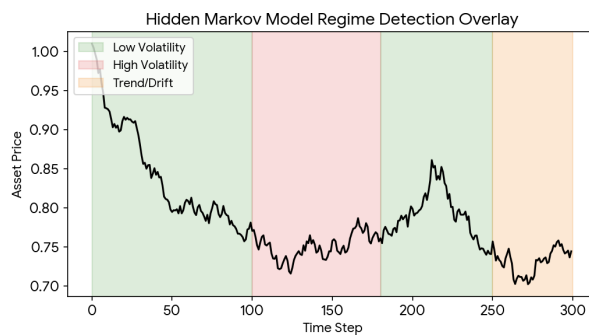


Figure 6: Regime Detection Overlay.

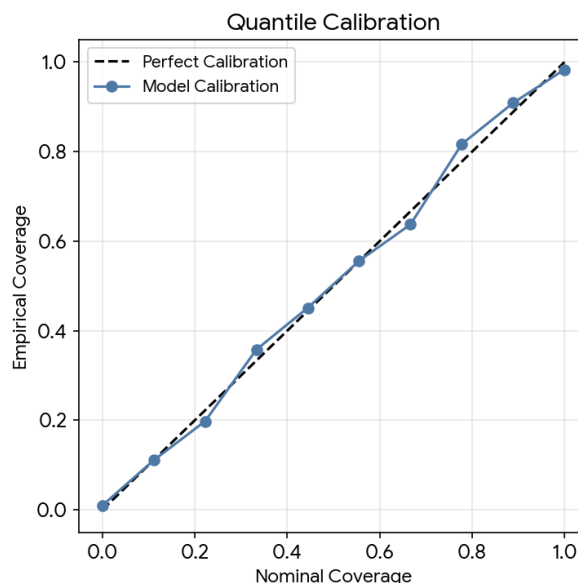


Figure 7: Quantile Calibration Curve.

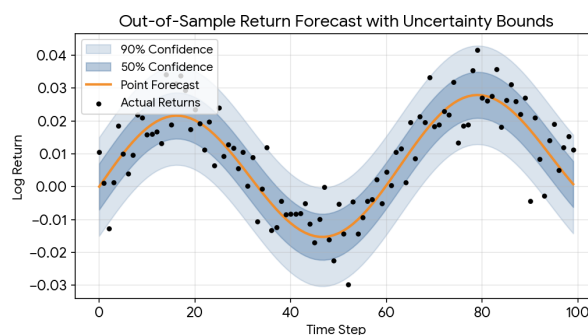


Figure 8: Forecast Uncertainty Fan.